

# SAMSON JOSE J

## Portfolio

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## PERSONAL SUMMARY

Computer Science (AI & ML) undergraduate graduating in 2027 with an 8.6 CGPA and strong foundations in Python, Data Structures & Algorithms, SQL, DBMS, and software development. Hands-on experience developing Python-based applications, integrating REST APIs, processing real-time data, debugging software, and using Git/GitHub for version control. Experienced in machine learning, data preprocessing, model evaluation, and end-to-end project development through academic projects and a Full Stack Web Development internship. Interested in Python development, automation, data processing, AI/ML, and modern software engineering practices.

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## EDUCATION

**ATME College of Engineering, Mysuru**

**Sept 2023 - (2027)**

Bachelor of Engineering in CSE (AI & ML) | **CGPA: 8.6**

**St. Philomena's PU College, Mysuru**

**Mar 2023**

PU in PCMC | **Percentage: 91%**

**St. Matthias High School, Mysuru**

**May 2021**

KSEEB | **Percentage: 73.6%**

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## TECHNICAL SKILLS

- **Programming:** Python, C++, C
- **Core CS:** Computer Networks, Data Structures & Algorithms, DBMS, Operating Systems
- **Software Development:** REST APIs, API Integration, Debugging, SDLC, Version Control
- **Data & AI/ML:** Pandas, NumPy, Scikit-Learn, TensorFlow, Data Preprocessing, Model Evaluation
- **Databases:** SQL, MongoDB
- **Web Technologies:** HTML
- **Developer Tools:** Git, GitHub, VS Code, Jupyter Notebook, Google Colab
- **Automation & Deployment:** GitHub Actions, Render

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## PROJECTS

### Brain Tumor Recognition Using Machine Learning | Ongoing

- Developing a Python-based machine learning system for brain tumor recognition using medical imaging data.
- Performing image preprocessing, feature analysis, model training, and evaluation for tumor recognition.
- Applying machine learning techniques to classify medical images and analyze prediction performance.
- Evaluating model performance and prediction reliability using appropriate classification metrics.

### Air Quality Index Prediction using Regression Real-Time | Mini Project

- Collected and preprocessed real-time AQI data from 94+ Indian cities using OpenWeather and IQAir APIs.
- Trained and compared Linear Regression, Random Forest, and XGBoost models, achieving up to 89% accuracy.
- Automated real-time data collection and periodic model retraining using GitHub Actions and deployed the workflow on Render.
- Developed a prediction pipeline to forecast AQI for the next 1–48 hours and generate health-related alerts.

### Anti-Spoof AI Attendance System | 24-Hour Hackathon

- Developed an AI-powered attendance system using Python, Flask, OpenCV, PyTorch, and YOLOv8 for real-time face detection and anti-spoofing.
- Built a multi-stage computer vision pipeline using YOLOv8 for face detection, RetinaFace for face alignment, and AdaFace for 512-dimensional face embeddings and recognition.
- Integrated Flask REST APIs with Firebase Firestore and Firebase Storage for student enrollment, face embeddings, authentication, attendance logging, and report generation.
- Developed a web interface using HTML, CSS, and JavaScript with automated attendance marking, CSV reporting, and attendance analytics.

## INTERNSHIP EXPERIENCE

### Full Stack Web Development Intern | Future Interns | 1 Month (Remote)

- Developed responsive web application components and implemented basic API integrations.
  - Worked across frontend and backend development while debugging application components.
  - Used Git/GitHub for version control and gained practical exposure to SDLC.
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## LEADERSHIP & ACHIEVEMENTS

### Event Coordinator | TechNova 2025 (IET Student Chapter)

- Planned and coordinated technical events for TechNova 2025 in collaboration with the organizing team.

### Student Placement Coordinator | Department of CSE (AI & ML)

- Coordinated placement-related activities between students and faculty
  - Supported students during placement preparation and technical events.
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## CERTIFICATIONS & ACADEMIC HONORS

### VTU BE Honors Programme – Visvesvaraya Technological University

- Programming in Java – 97%
- Introduction to Industry 4.0 and Industrial Internet of Things – 99%
- The Joy of Computing Using Python – 84%
- Introduction to Machine Learning – 85%

**NPTEL-** Programming in Python (82%)